

Ahsanullah University of Science and Technology

Department of Computer Science & Engineering

Course No: CSE 4174 | Course Name: Cyber Security Lab

Assignment 2

Topic: Audio Steganography

Task 1 — Record Your Name & Convert to WAV

Record yourself saying your full name using your phone. Upload the recording to Google Colab and run the converter code below.



python

```
# Install required libraries
```

```
!pip install pydub
```

```
!apt-get install -y ffmpeg
```

```
from pydub import AudioSegment
```

```
from google.colab import files
```

```
# Upload your recorded audio file
```

```
uploaded = files.upload() # upload your .m4a or .mp3 file here
```

```
# Get the uploaded filename
```

```
filename = list(uploaded.keys())[0]
```

```
# Convert to .wav
```

```
audio = AudioSegment.from_file(filename)
```

```
audio.export("my_recording.wav", format="wav")
```

```
print("Converted successfully! File saved as my_recording.wav")
```

Task 2 — Hide Your Student ID

Write a Python function to hide the **last 3 digits of your student ID** inside the .wav audio file using the LSB method. Save the output as encoded_audio.wav.

Hint — Useful Libraries:

- wave — built-in Python library for reading and writing .wav files
- struct — for handling binary data
- os — for file operations

Research how these libraries work and implement the encode function yourself.

Task 3 — Extract the Hidden ID

Write a Python function to extract the hidden digits from `encoded_audio.wav` and print them.

Hint: Use the same libraries from Task 2. Think about how you hid the bits — extracting is just the reverse process.

Submission

- Submit the Colab notebook (.ipynb) on Google Classroom
 - Bring a hardcopy of:
 - Your complete code
 - Step by step **handwritten explanation** of how your code works
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